

Towards Scalable Fault-Tolerant Quantum Computing Architectures through Time-Multiplexed Control



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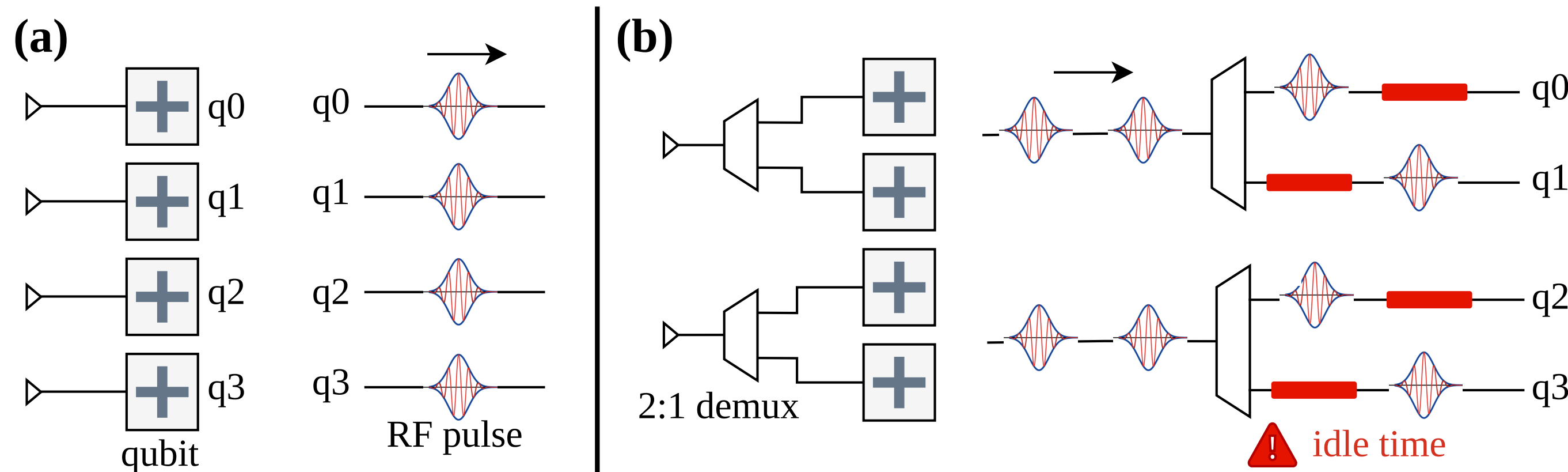
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Computer Architecture
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1. Motivation: I/O Wiring Bottleneck

- Scaling superconducting QPUs to the millions of qubits required for fault tolerance is bottlenecked by the number of cables routed into the dilution refrigerator.
- Time-Domain Multiplexing (TDM)** reduces wiring overhead but at the cost of multiplexed qubits idling while waiting on the shared channel and accumulating decoherence-induced errors.



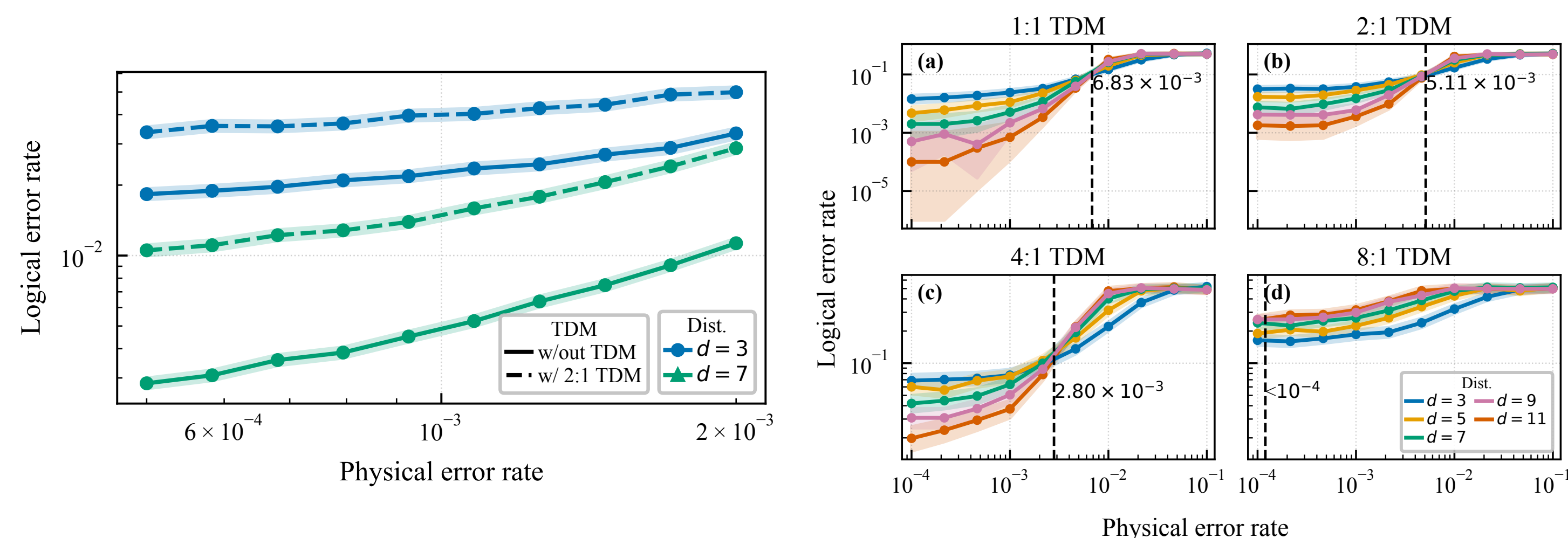
Research Question: Can surface-code error correction absorb the extra noise that TDM introduces, and where in the fault-tolerant stack does the cost concentrate?

2. Methodology

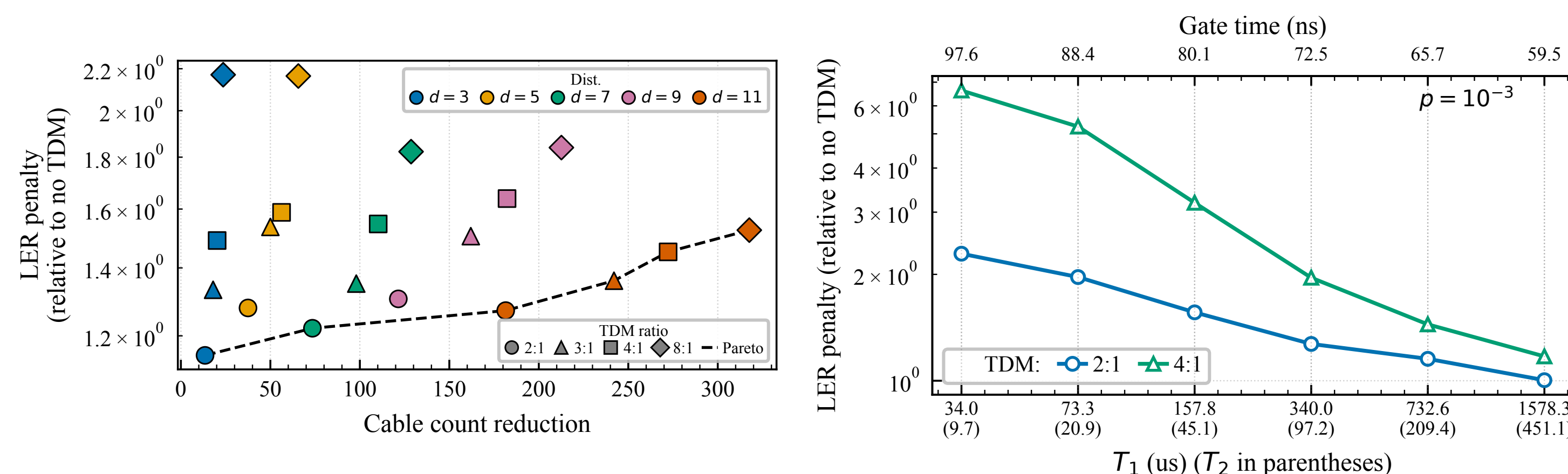
- Circuit-level Stim simulations of the rotated surface code. Amplitude/phase damping on idle qubits approximated w/ Pauli Twirling.
- Analytical evaluation of the Bravyi–Kitaev 5:1 T-state distillation protocol, extended to include amplitude- and phase-damping idling noise.
- Real (imperfect) demultiplexer-switches have finite isolation i which leaks a fraction $\lambda^2 = 10^{-i[\text{dB}]/10}$ of the drive pulse onto every idling (spectator) qubit of the switch. This results in gate infidelity which we approximate by a symmetric depolarizing channel applied to the spectator qubits.

3. Syndrome Measurement

LER degradation & threshold: TDM increases the logical error rate and shifts the threshold toward lower physical error rates.

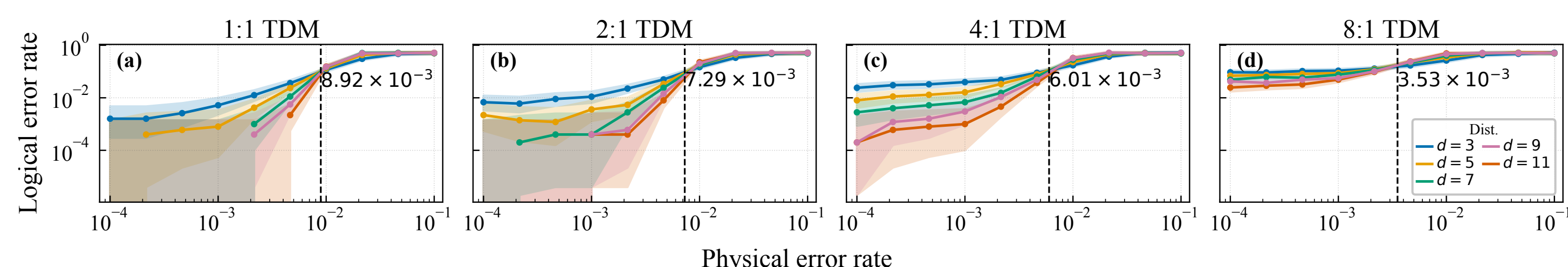


Tradeoff & hardware projections: Larger code distances better absorb the idling noise of TDM. As coherence times and gate speeds improve, the LER penalty of TDM is expected to decrease.



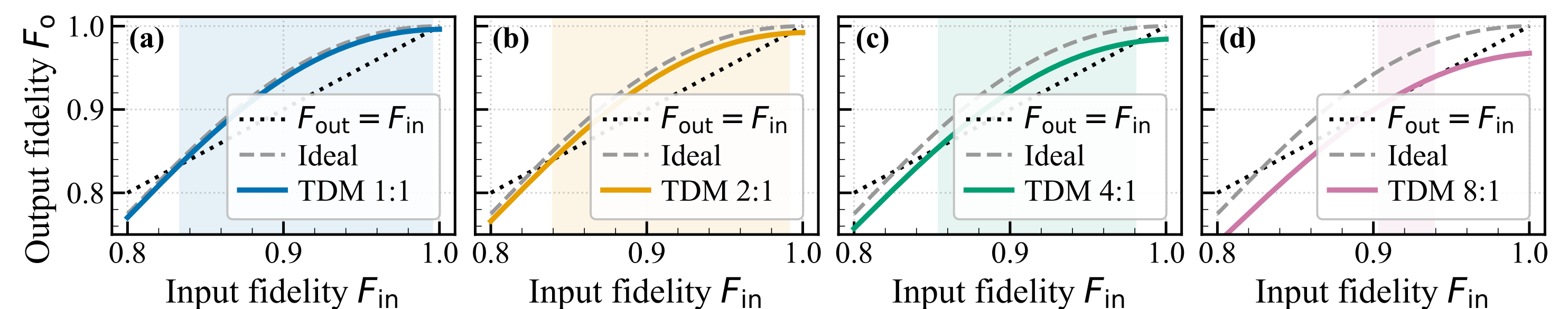
4. Lattice Surgery

Lattice Surgery Threshold: Additional idling noise from TDM impacts less merge-split operations threshold compared to syndrome measurement.



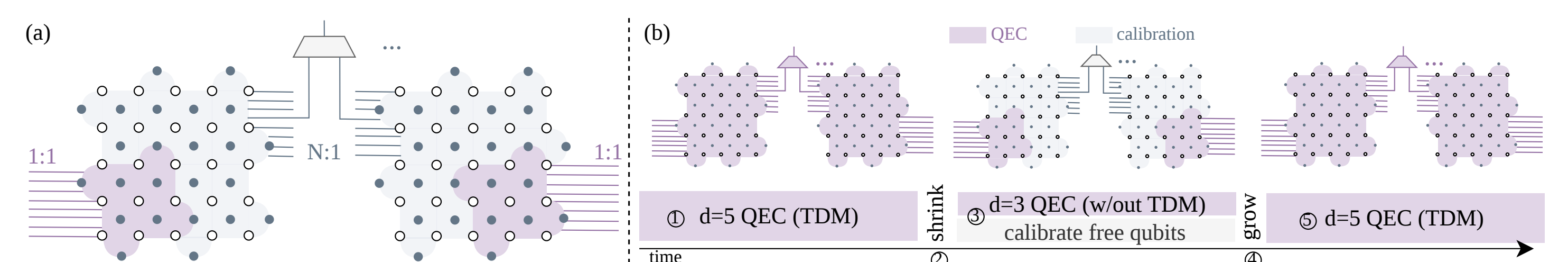
5. Magic State Distillation

Magic state fidelity: Magic state distillation takes several noisy input magic states and produces a smaller number of higher-fidelity output magic states and is only useful when the output fidelity exceeds the input fidelity (shaded region). Under TDM, this useful region shrinks as the multiplexing ratio grows.

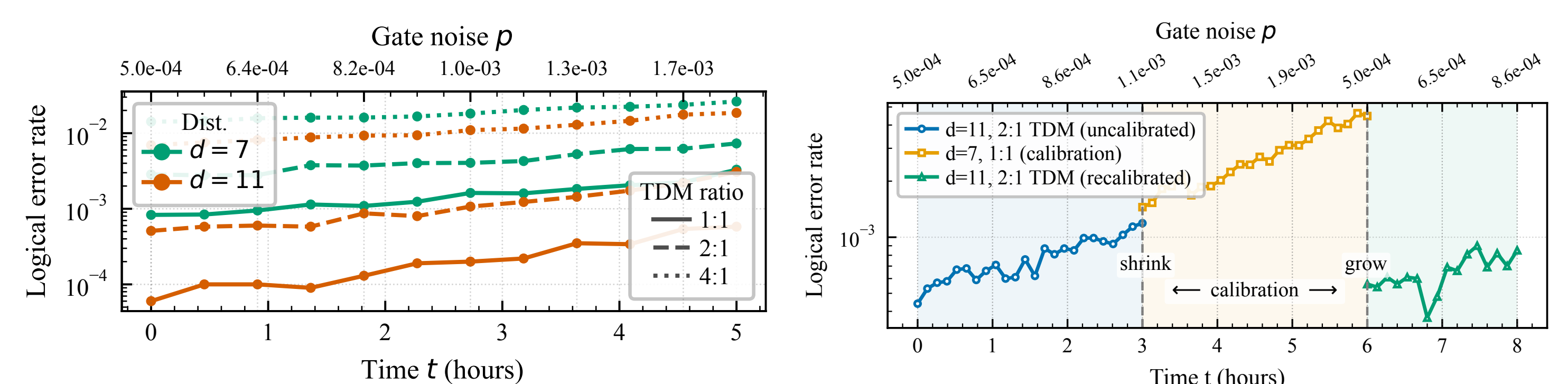


6. Multiplexing QEC with Recalibration

Can this existing TDM hardware be repurposed to also allow fault-tolerant execution to continue uninterrupted across recalibration events by multiplexing QEC with qubit recalibration routines?

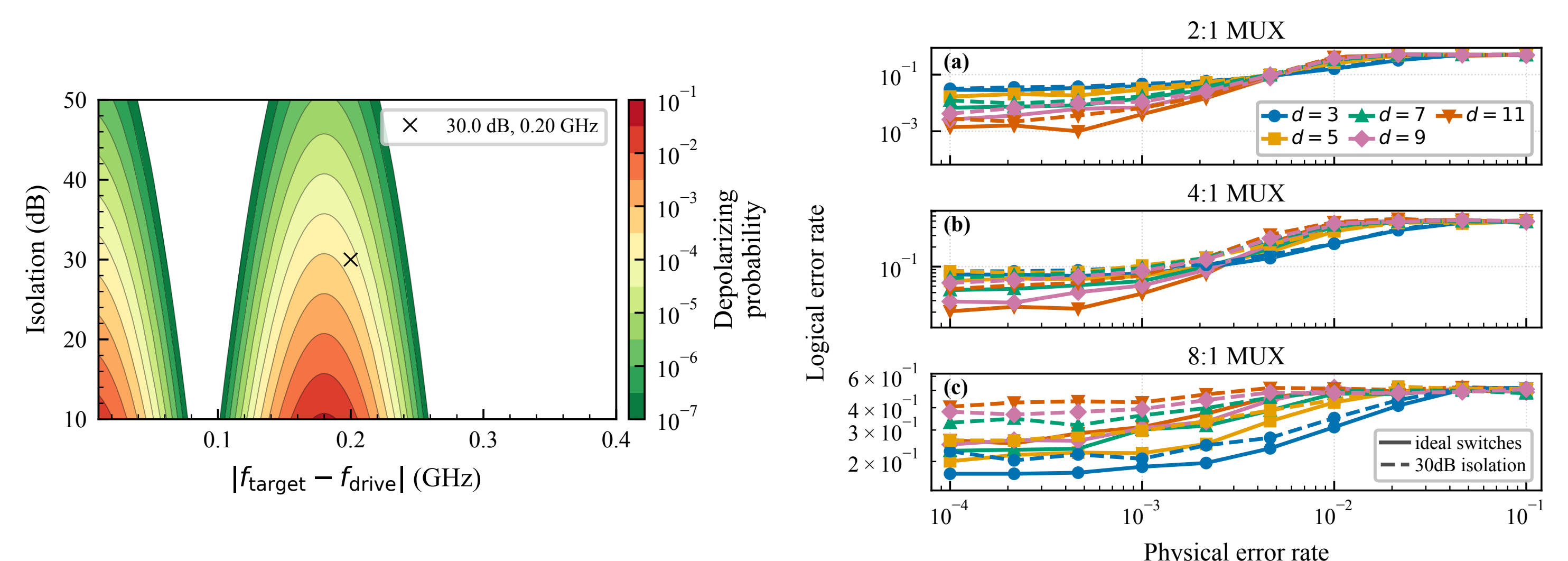


Hardware architecture: Partition each surface-code patch into an inner unmultiplexed sub-lattice (purple) and outer qubits (gray). When drift triggers recalibration (①), shrink the patch via code deformation (②), continue QEC unmultiplexed on the inner sub-patch and recalibrate the outer qubits (③) over the now-free multiplexed lines. Then grow back (④).



7. Imperfect Demultiplexer Switches

Depolarizing rate caused by finite port-to-port isolation vs. qubit-pulse detuning $\Delta/(2\pi)$ and switch isolation i and LER vs. physical error rate **for ideal switches and 30 dB isolation switches.**



8. Conclusions and Future Work

- The cost of TDM concentrates on *syndrome measurement*; *lattice surgery* is far less affected, *distillation* degrades gracefully only at moderate ratios.
- A 10 $\times$ improvement in coherence and gate times shrinks the LER penalty by two orders of magnitude, making future TDM essentially free.
- Aggressive multiplexing (i.e. 8:1) severely degrades output magic state fidelity and should be avoided in distillation factories.
- State-of-the-art cryogenic CMOS RF multiplexers (30 dB) already provide sufficient isolation. Therefore, the dominant cost of TDM is idling noise, not switch crosstalk.
- The same multiplexed control hardware can be reused to interleave *calibration* with QEC, for fault-tolerant execution not to pause for calibration.

Future work: Architectures for heterogeneous TDM (different ratios on data patches, ancilla patches, and distillation factories), co-designed with calibration.